

EquiPoint

Anshul Bhatt and Sara Hulyalkar

Problem Statement: Sickling

Sickling in a pointed foot

Fig 1

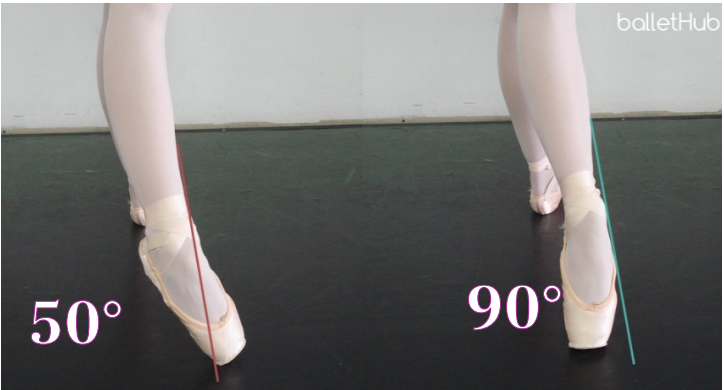


Fig 2



Source: <https://thedancephysio.com/banish-those-sickled-feet/>

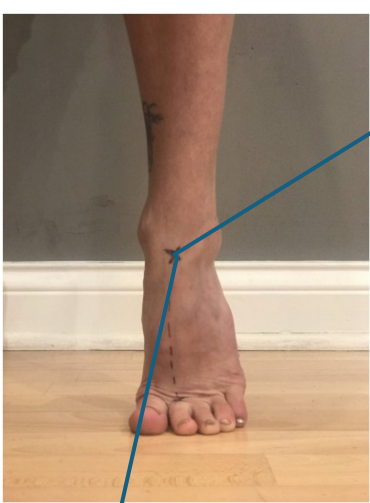
Fig 3



The angle on a correctly placed foot should be 90 degrees. A shift in this angle causes sickling

Sickling on demi-pointe or pointe (when dancing on one's toes or the tips of one's toes)

Fig 3



Second toe in line with the lateral malleolus

Source: <https://louettafootandankle.com/educational-resources/ballet-dancers-can-avoid-sickled-feet/>

Fig 4

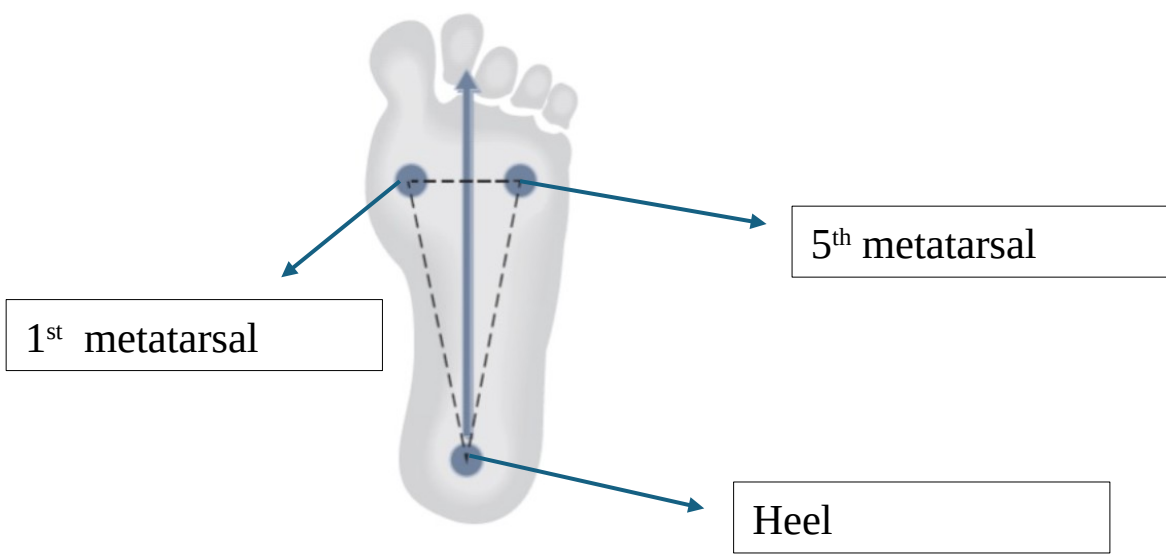


Second toe misaligned from the lateral malleolus

Source: <https://allegrodanceboutique.com/blogs/news/tips-to-prevent-sickling>

Problem Statement: Rolling and Weight Placement

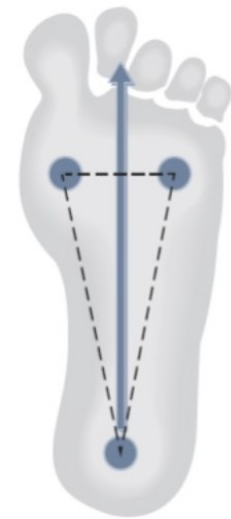
The Tripod Foot



While standing on the heel, 75% of the weight should equally be distributed between the 1st metatarsal, 5th metatarsal and 25% should be on the heel

While standing on the toes, weight shifts to the 1st and 5th metatarsals, with more pressure on the 1st metatarsal

Rolling



Excess weight on either the 1st or 5th metatarsal, causing the foot to 'roll' inwards or outwards

Rolling In



Fig 7: Diagram depicting rolling inwards

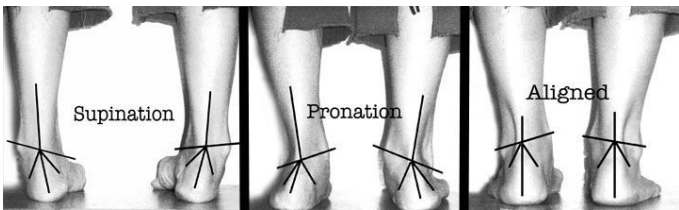


Fig 8: Diagram representing rolling inwards, outwards and an aligned foot from the heel

Problem Statement: Injury Analytics

Ballerinas frequently face issues with foot and toe alignment and weight distribution causing issues such as sickling, rolling and impair performance and increase the risk of injuries like ankle sprains and tendonitis. Improper weight distribution and alignment can have detrimental effects on dancers in the long run like making dancers more injury prone, joint degeneration, muscle imbalances and pain and reduced stability and balance, deeming the issue imperative to be solved.

- As seen in Figure 1, almost 41% of injuries in dancers are in the foot or the heel. Certain injuries in the knee, which makes up another 24% of injuries, can also be addressed in part by altering foot technique en pointe
- Sickling contributes to around 35% of all injuries while rolling contributes to 30%, making a total of 75% of injuries in dancers due to sickling and rolling alone
- Certain injuries due to overuse (dancing for prolonged periods of time without rest) can also be mitigated by improving foot technique and alignment

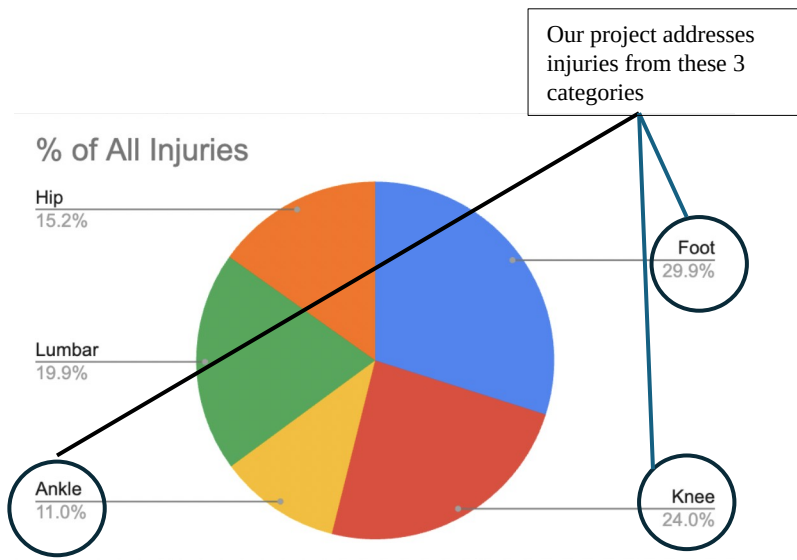


Fig 7: Pie chart depicting the % of total injuries in ballerinas in different parts of the body

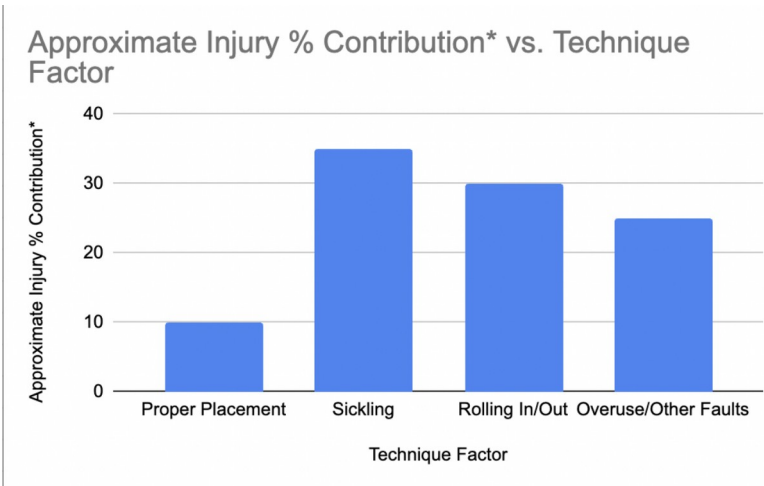
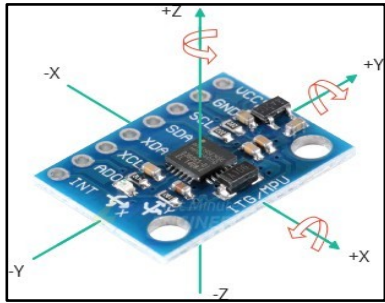


Fig 10: Bar graph showing the approximate injury percentage contribution for different technique factors

Injury	Associated Ballet Biomechanics	Main Symptoms and Early Signs
Achilles Tendinitis	Poor alignment and excessive plantarflexion	Heel/ankle pain, stiffness and swelling
Ankle Sprains	Rolling and sickling	Sudden pain, swelling and reduced mobility
Stress Fractures	Poor Weight distribution	Persistent localized pain, swelling, tenderness
Cuboid Subluxation	Rolling Inwards and Outwards	Lateral foot pain, sense of instability

Fig 11: a table depicting injuries and their associated ballet biomechanics along with symptoms

Methodology: Sensors and Data Processing Algorithms



MPU6050: Inertial Measurement Unit (IMU)

3D Motion Measurement: The MPU6050 is a sensor that measures acceleration in three axes (A_x, A_y, A_z) and angular velocity in three axes (G_x, G_y, G_z) as the dancer moves. The orientation of the sensor affects calibration and data obtained.

*Note. From MPU6050 Gyroscope and Accelerometer Module [Photograph], by LastMinuteEngineers.
<https://lastminuteengineers.com/mpu6050-accel-gyro-arduino-tutorial/>*

FSR402: Force Sensitive Resistor

The Force Sensitive Resistor (FSR) measures the normal force F applied by the dancer's foot. Its resistance R decreases nonlinearly with increasing force, typically following $R \propto F^{-n}$ (where $n \approx 0.5 - 1$).

When used in a voltage divider, the output voltage $V_{out} = V_{in} \frac{R_{fixed}}{R_{FSR} + R_{fixed}}$ provides a signal proportional to applied pressure $P = \frac{F}{A}$, enabling quantitative analysis of weight distribution and balance.

Madgwick Filter: Sensor Fusion Algorithm

With the IMU sensor, we use a Madgwick filter, which fuses gyroscope and accelerometer data to estimate a dancer's 3D orientation in real time. It minimizes the error between measured and expected gravitational field vectors using a gradient descent algorithm. The quaternion rate of change is given by

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes \boldsymbol{\omega} - \beta \nabla f(\mathbf{q}, \mathbf{a}, \mathbf{m})$$

where $\mathbf{q}$ is the orientation quaternion, $\boldsymbol{\omega}$ the angular velocity from the gyroscope, and β a gain parameter controlling convergence speed. The correction term $\nabla f(\cdot)$ drives the estimated direction of gravitational field to align with the measured ones, reducing drift without requiring matrix Jacobians or high computational cost.

After correcting the sensor outputs using this filter, we also log the four raw quaternions (q_0, q_1, q_2, q_3) since they are then used for risk-analysis based on orientation

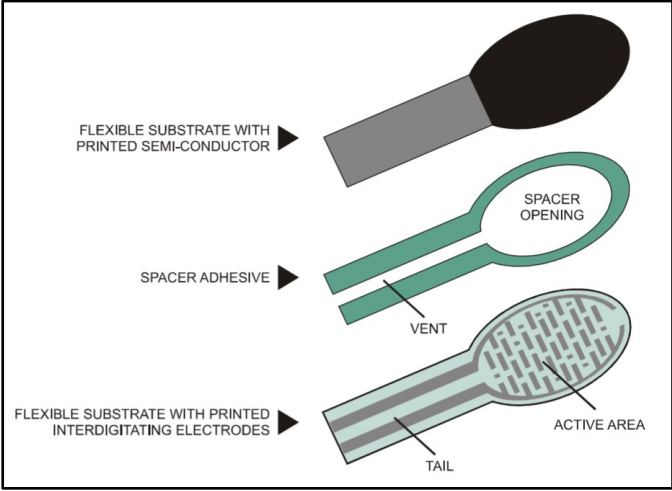


Fig 12
Diagram of
Force-
Sensitive
Resistor from
<https://makeabilitylab.github.io/physcomp/arduino/force-sensitive-resistors.html>

Methodology: Sensor Placement

According to research by Manna et al and Shih et al. the sensors were placed in the following areas on the shoe:

- FSR sensors under the heel and below the 1st and 5th metatarsals
- MPU 6050 placed on the dorsum (top) of the foot

FSR sensors were placed in this way because these regions are maximally impacted by sickling, rolling and improper weight distribution, ensuring the most sensitive location for injury prediction and feedback. The gyroscope sensor was placed in this way to capture angular changes related to inversion and alignment shifts.

Sensor	Position	Coordinates (m)	Notes
S1	Top-Left	(-0.060 , 0.065)	Near top left corner
S2	Top-Right	(0.060 , 0.065)	Near top right corner
S3	Center-Upper	(0.000 , 0.020)	Slightly above the center
S4	Bottom-Left	(-0.035, -0.045)	Lower left area
S5	Bottom-Right	(0.045, -0.065)	Lower right area

Fig 13: table indicating detailed FSR sensor placements

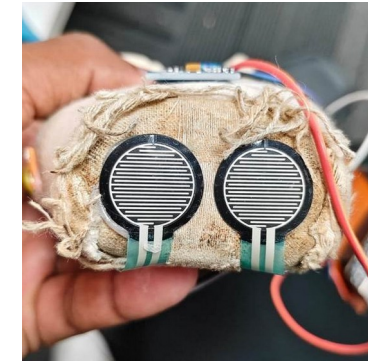


Fig 14 and 15: images showing the exact FSR sensor placements on the shoe

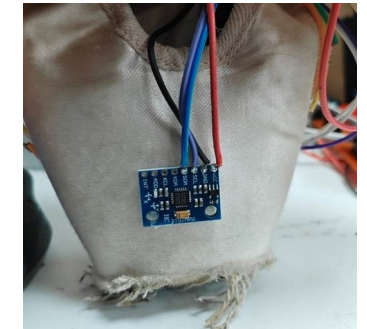


Fig 16 showing the exact MPU 6050 placement on the shoe

Methodology: Enclosures and Prototypes

Fig 3: Diagram of the 3D printed enclosure with labels

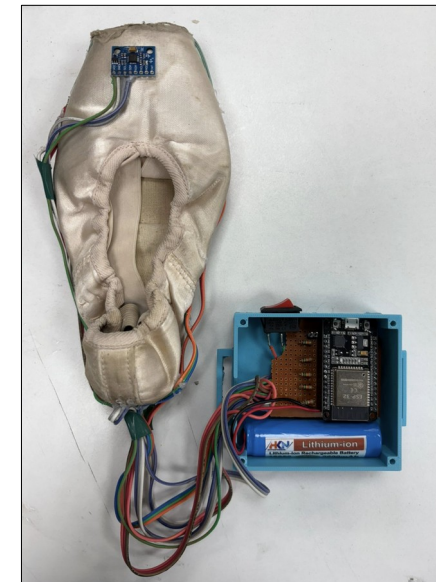
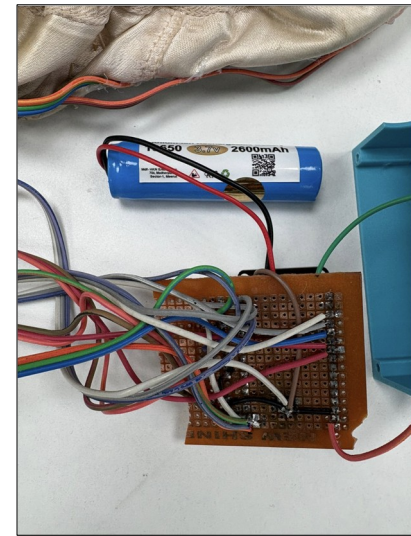
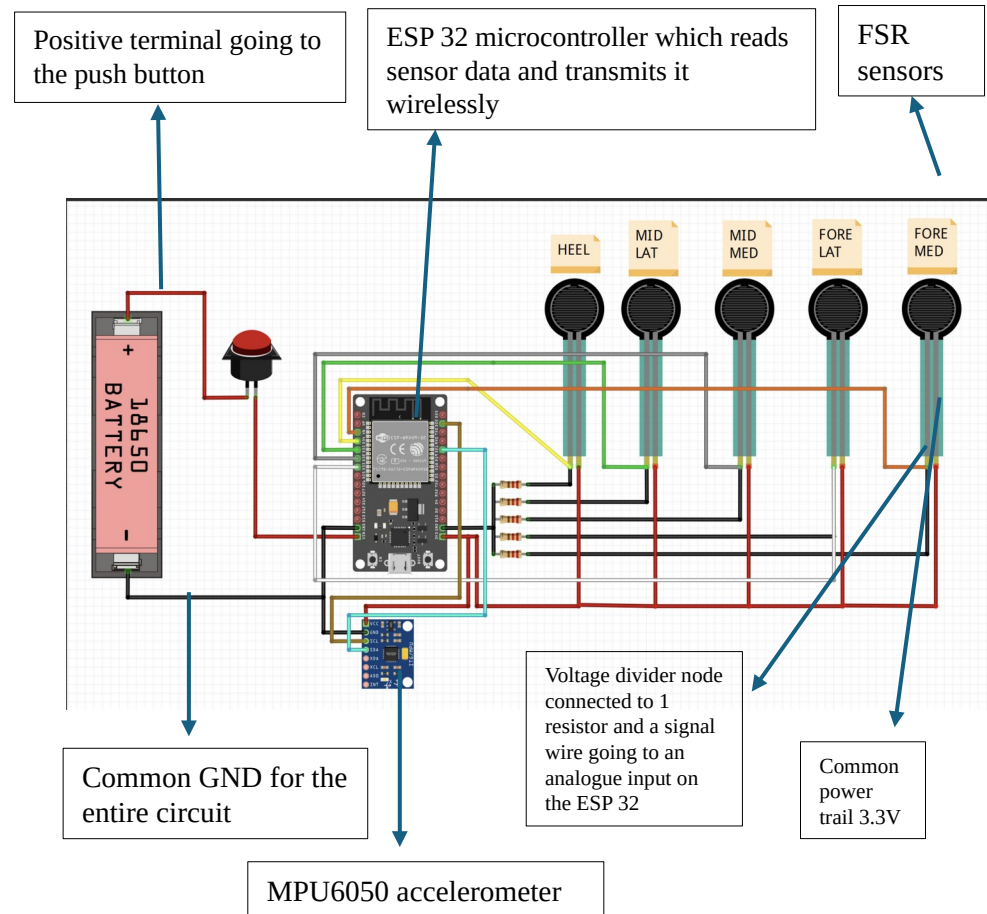
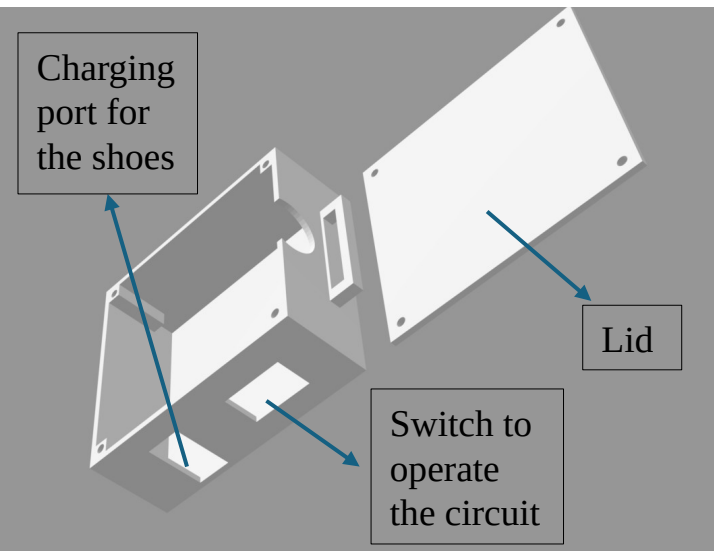


Fig 17 and 18:
Images of the circuit
with the shoe

Methodology: Risk Analysis

Sickling Detection: IMU + Toe FSR

Inputs:

IMU: Quaternion orientation $q = (q_0, q_1, q_2, q_3)$, Angular velocity ω_x & Vertical acceleration a_y .

Toe FSRs: S_1 (toe-left) & S_2 (toe-right)

The *roll angle* γ (foot tilt) is obtained from the IMU quaternions:

$$\gamma = \tan^{-1} \left(\frac{2(q_0q_1 + q_2q_3)}{1 - 2(q_1^2 + q_2^2)} \right)$$

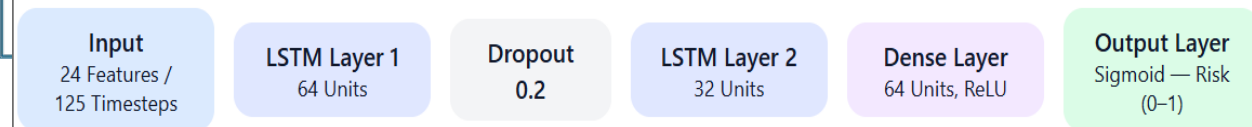
Toe pressure and imbalance are measured as:

$$P_{toe} = S_1 + S_2, \Delta_{toe} = \frac{S_1}{S_1 + S_2}$$

- Large $|\gamma|$: excessive inward or outward tilt
- High $|\Delta_{toe}|$: uneven toe loading
- High P_{toe} : dancer on pointe or demi-pointe

When all three occur together, the system classifies the movement as **sickled**.

Sickling Detection — LSTM (IMU + Toe FSR)



Rolling Detection — FSR Array

Goal: Identify rolling or side-to-side imbalance in the foot.

Inputs:

Five FSR sensors: S_1, S_2, S_3, S_4, S_5

Medial and lateral pressures are defined as:

$$P_{medial} = S_1 + S_4, P_{lateral} = S_2 + S_5$$

The Rolling Index (RI) quantifies side imbalance:

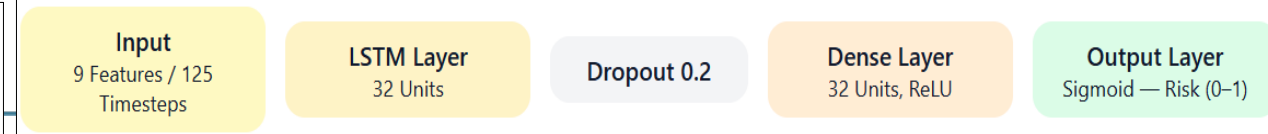
$$RI = \frac{P_{lateral} - P_{medial}}{P_{lateral} + P_{medial}}$$

Interpretation:

- $RI \approx 0$: balanced stance
- $RI > 0$: rightward roll
- $RI < 0$: leftward roll

An LSTM-based neural network tracks variations in RI and raw pressures over time, classifying each frame as Left Roll, Stable, or Right Roll, and producing a real-time rolling risk score between 0 and 1.

Rolling Detection — LSTM (FSR Array)



Methodology: App Interface

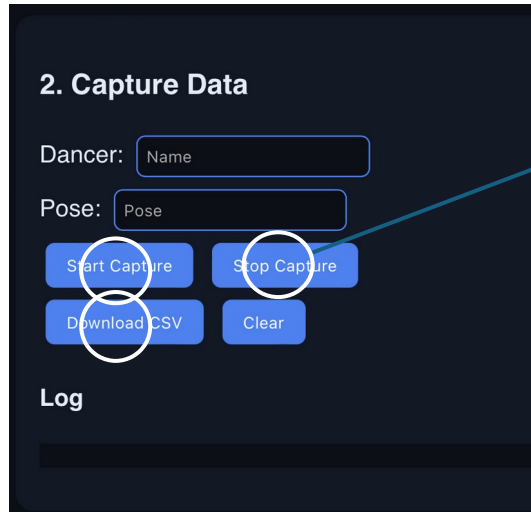


Fig 19

Left Shoe Capture Interface

Options to start and stop data collection

CSV files to download the data captured by the system

Options to start and stop risk analysis

Right and left toe calibration

Live risk analysis

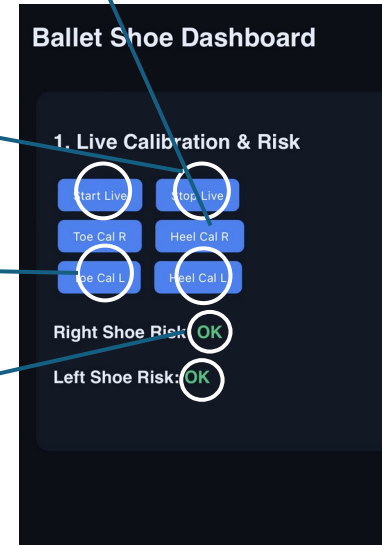


Fig 20

Right Shoe Capture Interface

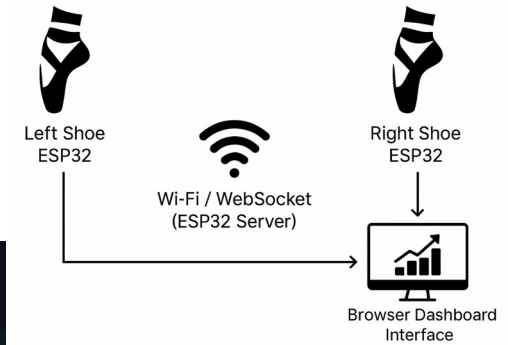
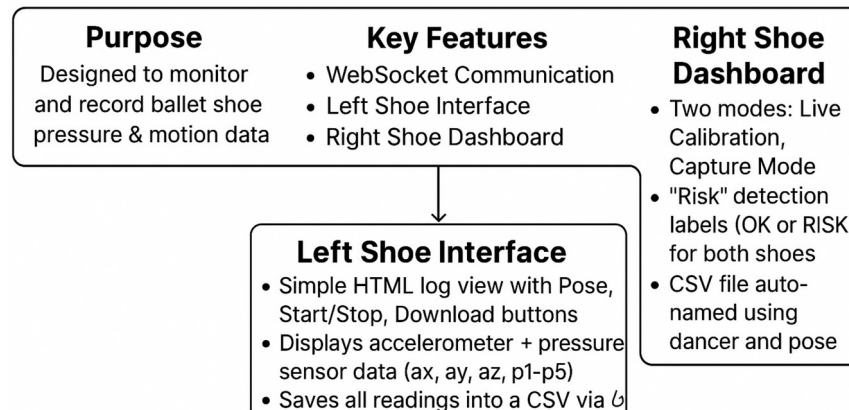


Diagram representing how the shoes are connecting to the interface



WEB TRANSMISSION

Left Shoe:

- Simple HTML log with Pose, Start/Stop, Download buttons
- Displays accelerometer + pressure sensor data (ax, ay, az, p1-p5)
- Saves all readings in a CSV file

Right Shoe:

- Two modes: Live calibration, Capture Mode
- Risk analysis detection: labels (OK or RISK) for both shoes
- CSV file auto-named using dancer and pose



Wifi/Websocket
(ESP 32 Server)



Browser Dashboard

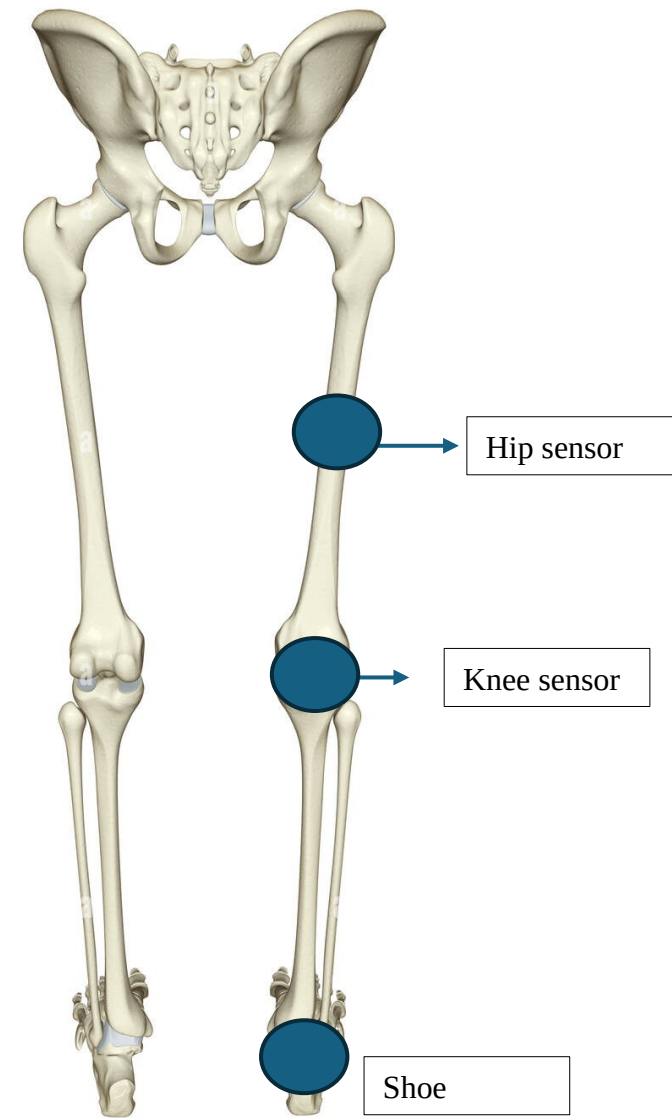
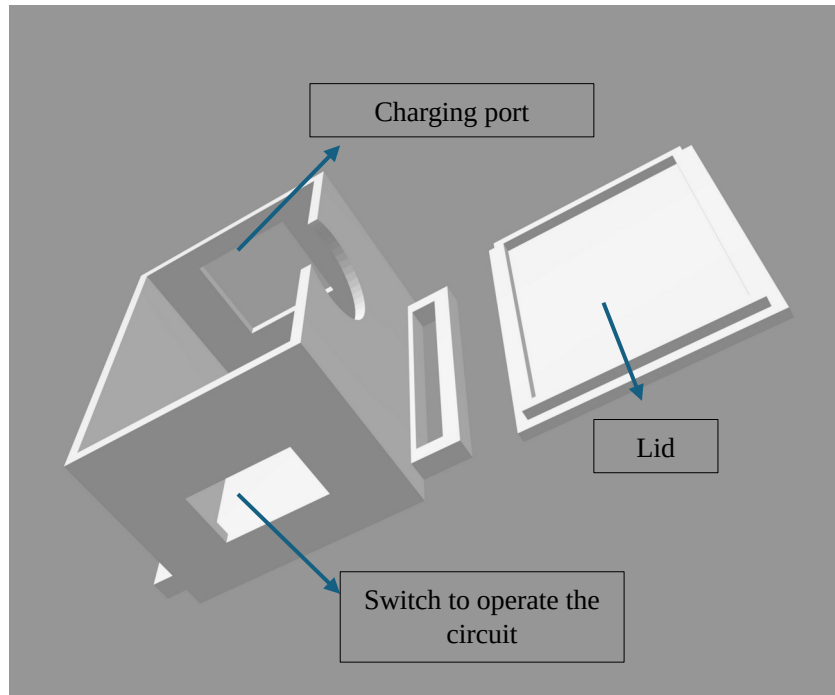
Ballet Shoe Dashboard

1. Live Calibration & Risk



Right Shoe Risk: **OK**

Left Shoe Risk: **OK**



Results

- A professional ballet instructor observed a dancer performing multiple relevé and plié movements while wearing the smart ballet shoe.
- The dancer was instructed to intentionally sickle or roll in certain trials to test the system’s ability to recognize incorrect technique.
- For each trial, the shoe’s sensor-based algorithm classified the movement as either normal or sickling/rolling, while the instructor independently recorded their own assessment.
- The collected data comparing the shoe’s predictions with the instructor’s evaluations were used to generate confusion matrices.
- These matrices show how accurately the shoe detected improper foot alignment compared to expert observation.

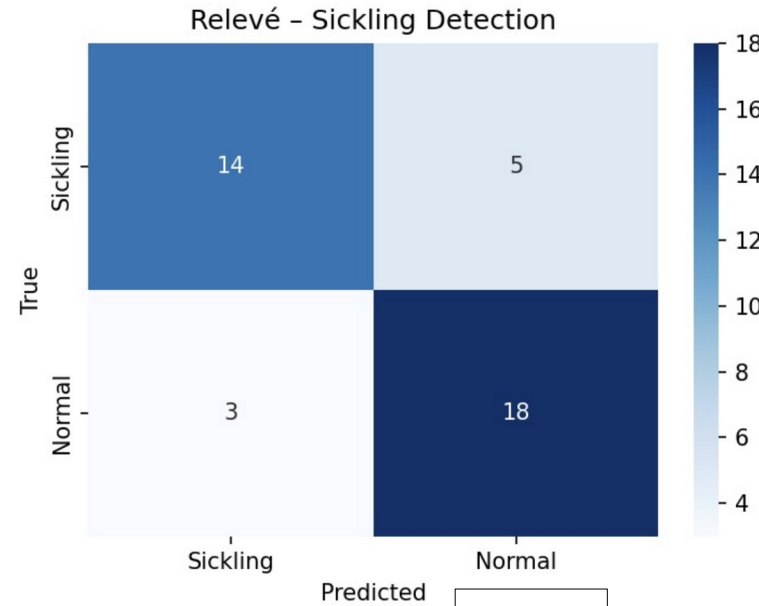


Fig 20

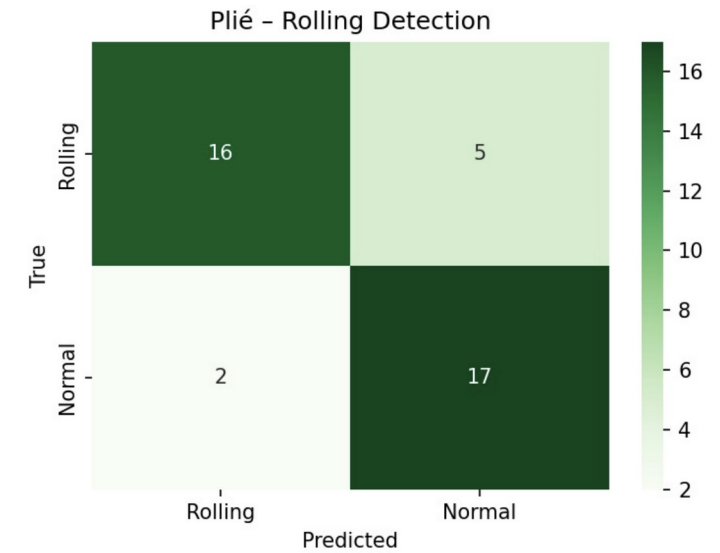


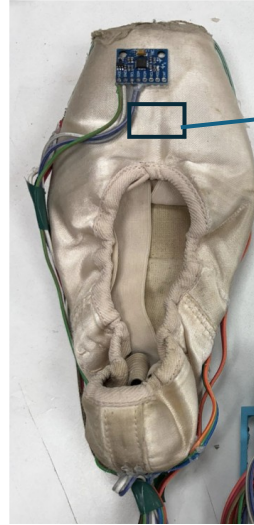
Fig 21

Task	Accuracy	Precision	Recall	Specificity	F1 Score
Relevé – Sickling Detection	80.0%	82.4%	73.7%	85.7%	77.8%
Plié – Rolling Detection	82.5%	88.9%	76.2%	89.5%	82.1%

Conclusions and Future Scope

We intend on generalizing the shoe to different sports where there is scope of such a shoe. E.g: Squash. Flexible wearable pressure sensors have revolutionized performance monitoring in a wide array of sports applications, enabling the evaluation of kinetic and physiological indicators across diverse activities such as running, tennis, rowing and more (Liu and Zhang)

Fig 22



Introduction of a second IMU sensor on the shoe to measure lines more accurately

In the future, we would like to improve our neural network architecture, including ResNet and bidirectional LSTMs. This is an attempt to improve classification and analysis of complex human movements in sports and dance.

We also plan on collecting more data across different ballet schools to get a more comprehensive understanding of the strengths and weaknesses of our product to improve on it. We intend on presenting the product to specialist ballet physiotherapists to get a better understanding of what the product may include.

Wearable sensors like ours enable real time monitoring of recovery, motor function and rehabilitation progress. Personalised physiotherapy based on data from the product enables active treatment and immediate adjustments.

Appendix: Quaternion Based Orientation and Kalman Filtration

Directly using the angular velocity obtained from the gyroscope is ineffective: the data is passed through an **Madgwick Filter** to reduce noise and obtain an accurate estimation of orientation. Then this is converted to a quaternion to portray orientation without **gimbal lock**, a major issue with **Euler angles** (another orientation quantification tool).

Note: **Madgwick** avoids the need to linearize the system using Jacobians, unlike the **Extended Kalman Filter**, which can introduce errors for highly non-linear systems.

$$\omega_q = [0, \omega_x, \omega_y, \omega_z]^\top$$

$$q = [q_w, q_x, q_y, q_z]^\top \quad \dot{q} = \frac{1}{2} q \otimes \omega_q$$

$$\begin{aligned} q_w &= \cos(\alpha/2) \\ q_x &= \sin(\alpha/2) \cos(\beta_x) \\ q_y &= \sin(\alpha/2) \cos(\beta_y) \\ q_z &= \sin(\alpha/2) \cos(\beta_z) \end{aligned}$$

$$\begin{aligned} \dot{q}_w &= \frac{1}{2} (-q_x \omega_x - q_y \omega_y - q_z \omega_z) \\ \dot{q}_x &= \frac{1}{2} (q_w \omega_x + q_y \omega_z - q_z \omega_y) \\ \dot{q}_y &= \frac{1}{2} (q_w \omega_y - q_x \omega_z + q_z \omega_x) \\ \dot{q}_z &= \frac{1}{2} (q_w \omega_z + q_x \omega_y - q_y \omega_x) \end{aligned} \Rightarrow \dot{q} = \frac{1}{2} \begin{bmatrix} -q_x \omega_x - q_y \omega_y - q_z \omega_z \\ q_w \omega_x + q_y \omega_z - q_z \omega_y \\ q_w \omega_y - q_x \omega_z + q_z \omega_x \\ q_w \omega_z + q_x \omega_y - q_y \omega_x \end{bmatrix}$$

$$q_{k+1} = q_k + \dot{q}_k \Delta t$$

$$k_1 = \dot{q}(q_k) \Rightarrow q_{k+1} = q_k + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_2 = \dot{q}(q_k + \frac{\Delta t}{2} k_1)$$

$$k_3 = \dot{q}(q_k + \frac{\Delta t}{2} k_2)$$

$$k_4 = \dot{q}(q_k + \Delta t \cdot k_3)$$

$$|q| = \sqrt{q_w^2 + q_x^2 + q_y^2 + q_z^2}$$

$$q_{\text{normalized}} = \begin{bmatrix} q_w/|q| \\ q_x/|q| \\ q_y/|q| \\ q_z/|q| \end{bmatrix}$$

$$q_1 \otimes q_2 = \begin{bmatrix} w_1 w_2 - x_1 x_2 - y_1 y_2 - z_1 z_2 \\ w_1 x_2 + x_1 w_2 + y_1 z_2 - z_1 y_2 \\ w_1 y_2 - x_1 z_2 + y_1 w_2 + z_1 x_2 \\ w_1 z_2 + x_1 y_2 - y_1 x_2 + z_1 w_2 \end{bmatrix}$$

Quaternion Multiplication
(non-commutative & binary)

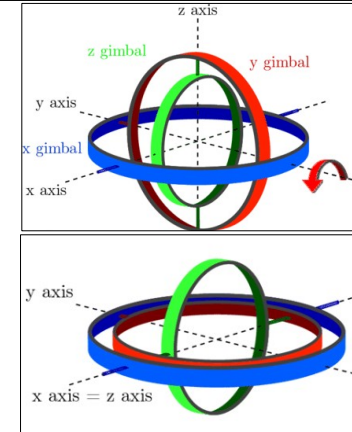


Fig.6: Visualising Gimbal Lock

Rotation matrix for rendering orientation in 3D

$$R = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_w q_z) & 2(q_x q_z + q_w q_y) \\ 2(q_x q_y + q_w q_z) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_w q_x) \\ 2(q_x q_z - q_w q_y) & 2(q_y q_z + q_w q_x) & 1 - 2(q_x^2 + q_y^2) \end{bmatrix}$$

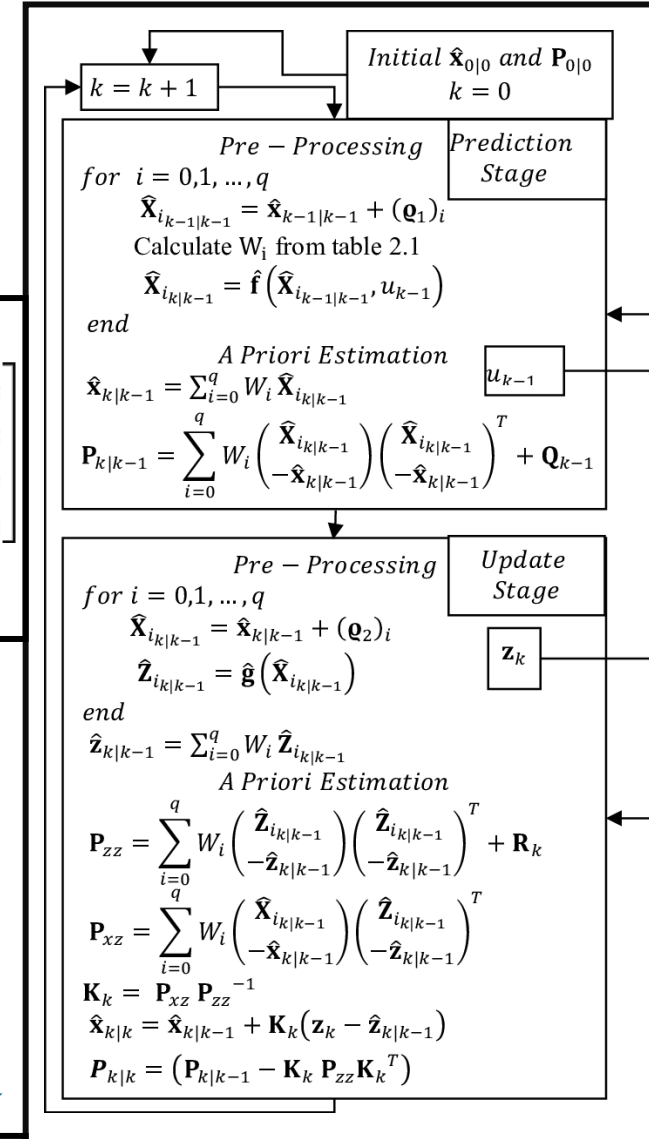


Fig.7: Unscented Kalman filter:
Mathematical approaches
(ResearchGate)

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Ballet Shoe Dashboard

1. Live Calibration & Risk

Start Live

Stop Live

Toe Cal R

Heel Cal R

Toe Cal L

Heel Cal L

Right Shoe Risk: **OK**

Left Shoe Risk: **OK**